

v0.2.0 · case study · 2026

Self-running social-video pipeline.

From campaign config to published Reels and TikToks — autonomously. State-machine in Postgres, stateless TypeScript workers, n8n orchestration, live dashboard with HITL approval gates that flip per-campaign to fully autonomous.

CODE

4,000+

lines of TS

WORKERS

11

stateless

STATES

13

audit logged

DEMO COST

\$0

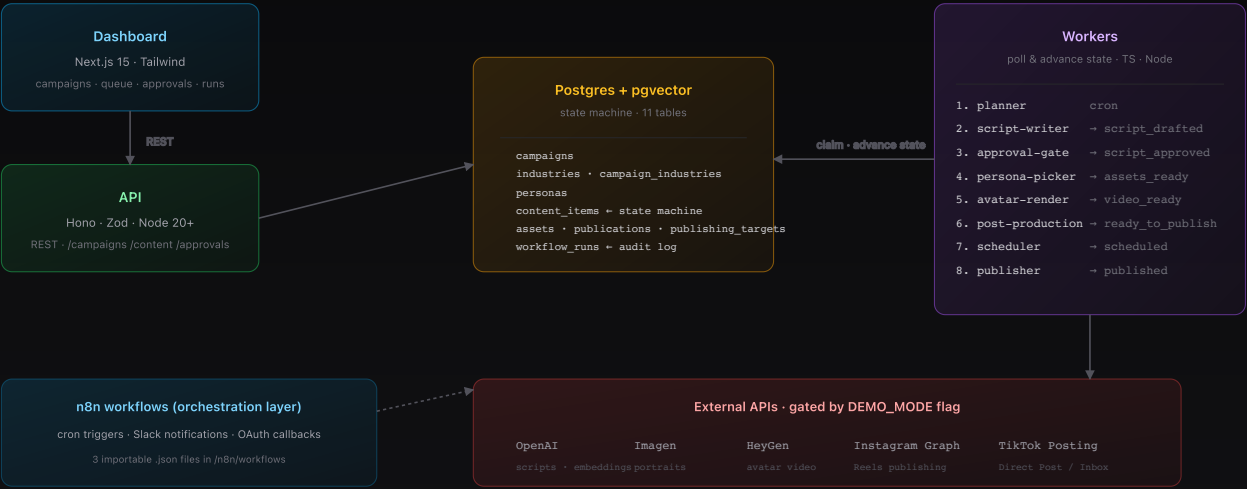
DEMO_MODE

ARCHITECTURE

Boring tech, tight integration

social-agent — architecture

State-machine in Postgres · stateless workers · HITL → auto autonomy modes



docker compose up · npm install · npm seed · npm devall → http://localhost:3000

Postgres owns business state – stateless workers consume it.

LIVE DASHBOARD

Every transition visible

social-agent

OverviewCampaignsQueueApprovalsRuns

demo mode

Overview

Pipeline health across all campaigns.

PLANNED

0

awaiting script

IN FLIGHT

6

workers active

SCHEDULED

1

publish queue

PUBLISHED

26

live on platforms

FAILED

0

needs review

Campaigns

Manage

CAMPAIGN	MODE	SCHEDULE	PLATFORMS	WEEKLY TARGET
Demo Brand Portfolio demo: a fictional growth-marketing brand serving local businesses across multiple verticals.	hitl	0 9,17 * * * · Europe/Berlin	📷 🎵	33 /wk

State distribution

Published

26

Awaiting approval

6

Scheduled

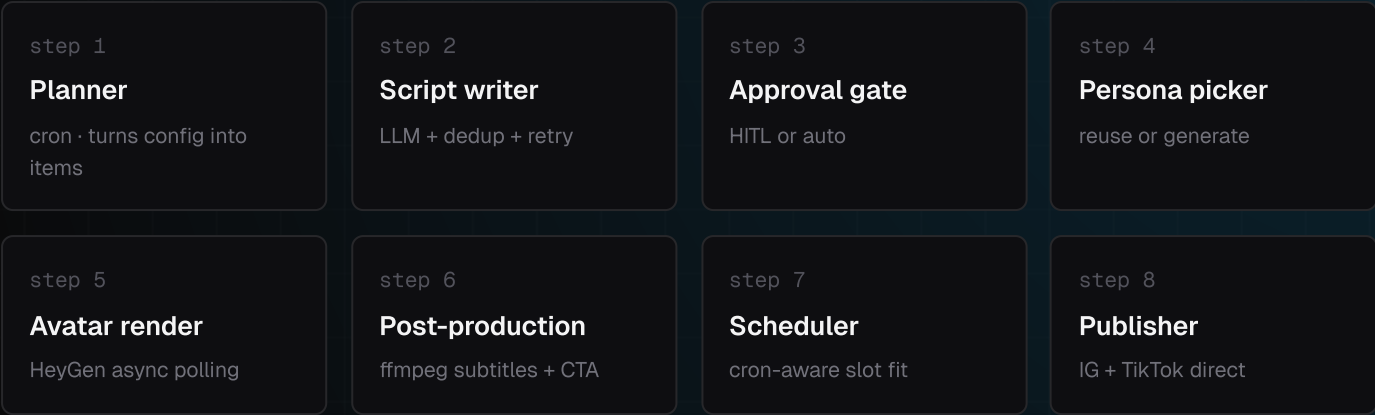
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N

Overview · pipeline health across campaigns

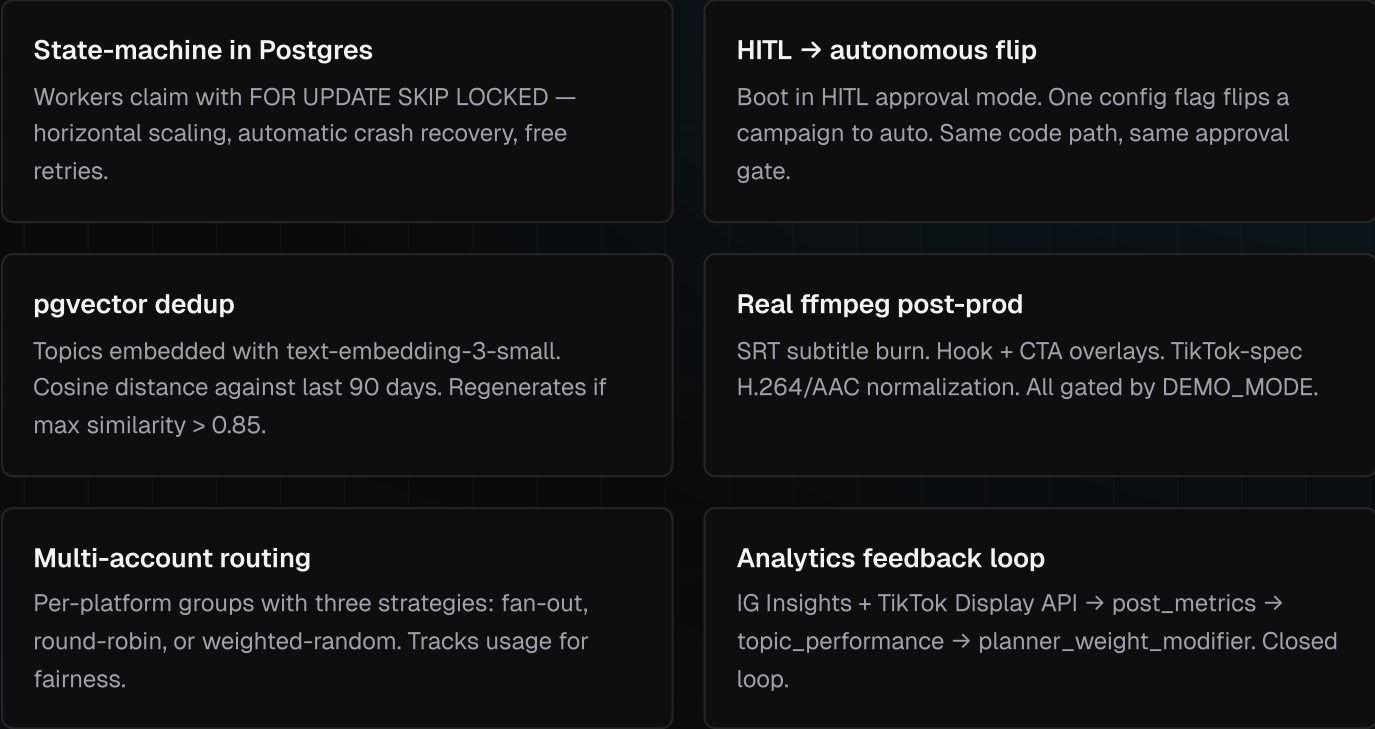
PIPELINE

Eight workers, one shared state



ENGINEERING HIGHLIGHTS

What's interesting under the hood



STACK

Production-credible primitives

- **Postgres 16 + pgvector + uuid-oss + citext + pg_trgm** — 11 tables, 13-state strict-forward state machine, 1536-dim embeddings with ivfflat ANN index
- **TypeScript monorepo (pnpm workspaces)** — services/core (Drizzle ORM, providers), services/workers, services/api (Hono), dashboard (Next.js 15)
- **n8n for orchestration** — three importable workflows for cron, Slack approval routing, publishing health monitor
- **External integrations** — OpenAI scripts/embeddings, HeyGen avatar render, Google Imagen portraits, Pexels B-roll, IG Graph API + TikTok Direct Post — every one gated behind DEMO_MODE with a mock implementation that ships realistic data with realistic delays

QUICKSTART

Three commands to running pipeline

